

Cloud Security Assessment Report

GCP Firewall Security Lab: Secure Remote Access Implementation

Author

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Role

Future Cloud Security Engineer

Platform

Google Cloud Platform (GCP)

Assessment Type

Cloud Security Control Validation

Assessment Date

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Section 1: Identify the Cloud Security Control Implemented

The primary security control implemented during this assessment was a Google Cloud Platform (GCP) VPC Firewall Rule designed to restrict Secure Shell (SSH) administrative access to a Compute Engine virtual machine.

The objective of the control was to reduce the attack surface of the cloud workload by limiting inbound SSH traffic (TCP Port 22) to a single trusted public IP address instead of allowing unrestricted Internet access.

The assessment environment consisted of a Debian Linux virtual machine hosted within Google Cloud Compute Engine. A firewall rule named:

allow-ssh-from-my-ip

was configured to permit SSH traffic only from the administrator's authorized public IP address.

The firewall rule was applied using the network tag:

ssh-secure

which ensured that only specifically tagged virtual machines received the security policy.

This implementation follows cloud security best practices including:

- Principle of Least Privilege (PoLP)
- Network Access Control
- Secure Administrative Access
- Attack Surface Reduction
- Cloud Infrastructure Hardening

The primary goal of this project was to validate that administrative access could be successfully restricted without impacting legitimate management activities.

Section 2: Document the Security Assessment

The assessment began with the creation of a dedicated Google Cloud project to isolate resources and maintain administrative control throughout the testing process.

The Compute Engine service was enabled and a Debian Linux virtual machine named:

secure-vm

was deployed within the Google Cloud environment.

During the deployment process, a network security strategy was developed to prevent unrestricted administrative access from the Internet. Instead of relying on default access settings, the assessment implemented a dedicated firewall policy that restricted inbound SSH traffic.

The assessor identified the trusted public IP address used for administration and configured a VPC firewall rule to permit SSH traffic exclusively from that source. The firewall policy was configured with the following parameters:

Protocol: TCP

Port: 22

Direction: Ingress

Action: Allow

Source: Trusted Public IP Address (/32)

Target: ssh-secure

After the firewall rule was created, the virtual machine was assigned the corresponding network tag, ensuring the security policy was applied only to the intended workload.

The effectiveness of the security control was validated through a series of administrative tests. Secure Shell (SSH) access was initiated from the trusted source IP address and a successful connection was established.

The following validation commands were executed:

```
hostname
```

```
whoami
```

```
ip addr
```

The results confirmed:

- Successful authentication
- Correct hostname configuration
- Valid user account access
- Operational network interface
- Proper internal IP assignment

The assessment confirmed that legitimate administrative access remained functional while network exposure was significantly reduced.

No unauthorized access paths were identified during testing.

Following validation, all cloud resources were securely removed to eliminate unnecessary costs and maintain a clean cloud environment.

Section 3: Recommend Security Improvements

Several additional security controls can further strengthen the security posture of the cloud environment.

Multi-Factor Authentication (MFA)

Administrative accounts should be protected using MFA to reduce the likelihood of account compromise.

Even if an attacker obtains valid credentials, access would remain restricted without the second authentication factor.

Identity-Aware Proxy (IAP)

Google Cloud Identity-Aware Proxy can be used to eliminate direct SSH exposure and provide identity-based access control.

This approach significantly reduces the attack surface by removing direct inbound administrative connectivity.

Security Monitoring

Cloud Logging and Cloud Monitoring should be enabled to detect:

- Unauthorized access attempts
- Firewall modifications
- Configuration changes
- Suspicious authentication activity

Continuous monitoring improves visibility and supports incident detection.

Custom VPC Segmentation

Future implementations should utilize dedicated custom VPC networks and subnets rather than default networking configurations.

Benefits include:

- Improved segmentation
- Reduced lateral movement opportunities
- Enhanced security management

Least Privilege IAM

Cloud Identity and Access Management (IAM) roles should be assigned using least-privilege principles.

Users should receive only the permissions required to perform their job functions.

This minimizes risk associated with credential compromise and insider threats.

Assessment Outcome

The assessment successfully demonstrated the implementation and validation of cloud network security controls within Google Cloud Platform.

The configured firewall policy effectively restricted SSH access to a trusted source IP address while maintaining legitimate administrative functionality.

The project successfully applied industry-standard cloud security principles including:

• Principle of Least Privilege • Network Access Control • Administrative Access Restriction • Attack Surface Reduction • Secure Cloud Resource Management

The assessment confirms that properly configured VPC firewall rules provide an effective mechanism for protecting cloud-hosted workloads against unauthorized remote access attempts.

Executive Conclusion

This project demonstrates practical Cloud Security Engineering skills through the deployment, hardening, validation, and secure decommissioning of Google Cloud infrastructure.

By implementing IP-restricted SSH access, network tag-based policy enforcement, and least-privilege access controls, the environment achieved a significantly improved security posture while maintaining operational accessibility.

The skills demonstrated in this assessment align directly with responsibilities commonly performed by Cloud Security Engineers responsible for securing cloud infrastructure in enterprise environments.